

# RK3399 IO Domain Configuration Developer Guide

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ID: RK-SM-YF-906

Release Version: V1.0.0

Release Date: 2021-05-15

Security Level: ☐Top-Secret ☐Secret ☐Internal ☒Public

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## Preface

### Overview

The IO level of the controller's power domain must be matched with the IO level of the connected peripheral chip, and the voltage configuration of the software must be consistent with the voltage of the hardware. Otherwise, it may cause IO damage at worst.

This document mainly introduce the ways to configure IO power domain of RK3399 Linux SDK platform, aiming to help developers to configure IO power domain correctly.

### Product Version

| Chipset | System Version        |
|---------|-----------------------|
| RK3399  | Linux 4.4, Linux 4.19 |

### Intended Audience

This document (this guide) is mainly intended for:

- Technical support engineers
- Software development engineers
- Hardware development engineers

### Revision History

| Version | Author      | Date       | Change Description |
|---------|-------------|------------|--------------------|
| V1.0.0  | Caesar Wang | 2021-05-15 | Initial version    |

## Contents

### **RK3399 IO Domain Configuration Developer Guide**

1. Step 1: Obtain the Hardware Schematic Diagram and Check the Design of the Hardware Power Supply
2. Step 2: Find the Corresponding Kernel dts Configuration File
3. Step 3: Modify the Power Domain Configuration Node pmu\_io\_domains of the Kernel dts
4. Step 4: Check the Current Firmware IO Domain Configuration from SDK
5. Step 5: Confirm Whether the Register Value is Correct after Flashing the Firmware

## 1. Step 1: Obtain the Hardware Schematic Diagram and Check the Design of the Hardware Power Supply

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It will take RK\_IND\_EVB\_RK3399\_LP4D200P232SD8\_V13\_20200615 EVB as an example to introduce in this document.

Hardware schematic diagram is: RK\_IND\_EVB\_RK3399\_LP4D200P232SD8\_V13\_20200615.pdf.

Power solution: checking from the hardware schematic, the power solution of the **EVB RK\_IND\_EVB\_RK3399\_LP4D200P232SD8\_V13\_20200615** is with a PMU (RK809-3).

## 2. Step 2: Find the Corresponding Kernel dts Configuration File

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From the first step, it can be seen that the hardware power supply design of the EVB is with a PMU , and the corresponding kernel dts configuration file is located in:

arch/arm64/boot/dts/rockchip/rk3399-evb-ind.dtsi (The solution discussed in this document)

## 3. Step 3: Modify the Power Domain Configuration Node pmu\_io\_domains of the Kernel dts

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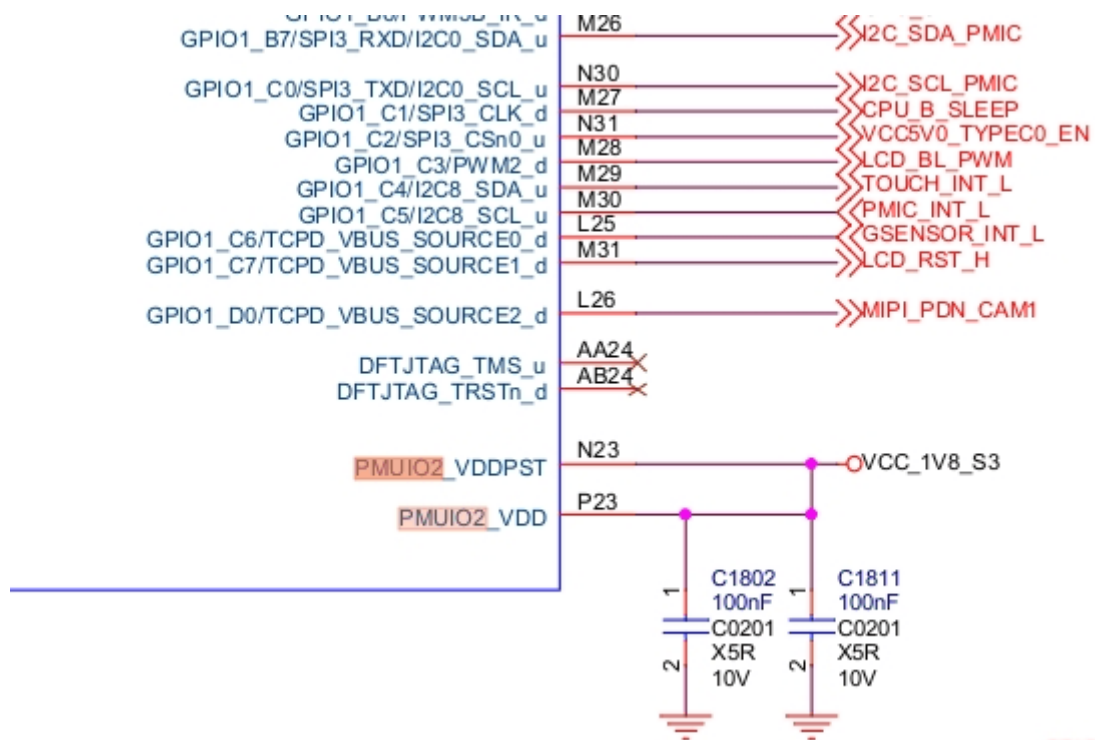
```
&io_domains {
    status = "okay";

    bt656-supply = <&vcc_3v0>;          /* bt656_gpio2ab_ms */
    audio-supply = <&vcca_1v8>;          /* audio_gpio3d4a_ms */
    sdmmc-supply = <&vccio_sd>;           /* sdmmc_gpio4b_ms */
    gpio1830-supply = <&vcc_3v0>;         /* gpio1833_gpio4cd_ms */
};

&pmu_io_domains {
    status = "okay";
    pmu1830-supply = <&vcc_1v8>;
};
```

Take **pmuio2-supply** for example, firstly, check the hardware schematic diagram to confirm the configuration of the pmuio2 power domain (PMUIO2) as shown in the below figure.

From the figure, you will find that the power supply of **PMUIO2** is VCC\_1V8\_S3 (that is 1.8V)



In the same way, you can find from the hardware schematic diagram that bt656-supply is connected to APIO2\_VDD, audio-supply is connected to APIO5\_VDD, and gpio1830-supply is connected to APIO4\_VDD.

## 4. Step 4: Check the Current Firmware IO Domain Configuration from SDK

Command: `./build.sh info`

```

请再次确认板级的电源域配置!!!!!!
esp 特别是M1-F1, FLASH, 以太网这几路电源的配置 >>>!!!!!!
检查内核文件 /home/wxt/linux-develop/rk3399/kernel/arch/arm64/boot/dts/rockchip/rk3399-cvb-1n3-lpddr4-linux.dts 的节点 [pmu_io_domains]

pmu1830-supply
regulator-min-microvolt = 1800mV
regulator-max-microvolt = 1800mV

bt656-supply
regulator-min-microvolt = 3000mV
regulator-max-microvolt = 3000mV

audio-supply
regulator-min-microvolt = 1800mV
regulator-max-microvolt = 1800mV

sdmmc-supply
regulator-min-microvolt = 1800mV
regulator-max-microvolt = 3300mV

gpio1830-supply
regulator-min-microvolt = 3000mV
regulator-max-microvolt = 3000mV

```

## 5. Step 5: Confirm Whether the Register Value is Correct after Flashing the Firmware

Take **RK3399** as an example, get PMU\_SOC\_CON0 register (0xFF320180) and GRF\_IO\_VSEL register (0xFF77E640) from the manual, they are shown as follows:

**PMUGRF\_SOC\_CON0**

Address: Operational Base + offset (0x00180)

SoC control register 0

| Bit | Attr | Reset Value | Description |
|-----|------|-------------|-------------|
|-----|------|-------------|-------------|

**RK3399 TRM**

| Bit   | Attr | Reset Value | Description                                                                                                                                                                                                                                                                                                                                                                         |
|-------|------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | RW   | 0x0000      | write_enable<br>bit0~15 write enable<br>When bit 16=1, bit 0 can be written by software .<br>When bit 16=0, bit 0 cannot be written by software;<br>When bit 17=1, bit 1 can be written by software .<br>When bit 17=0, bit 1 cannot be written by software;<br>.....<br>When bit 31=1, bit 15 can be written by software .<br>When bit 31=0, bit 15 cannot be written by software; |
| 15:10 | RO   | 0x0         | reserved                                                                                                                                                                                                                                                                                                                                                                            |
| 9     | RW   | 0x1         | pmu1830_vol<br>pmu IO 1.8v/3.0v select.<br>0: 3.0v ;<br>1: 1.8v ;                                                                                                                                                                                                                                                                                                                   |
| 8     | RW   | 0x1         | pmu1830_volsel<br>pmu GPIO1 1.8v/3.0v control source select.<br>0: controlled by IO_GPIO0B1 ;<br>1: controlled by<br>PMUGRF.SOC_CON0.pmu1830_vol                                                                                                                                                                                                                                    |
| 7     | RO   | 0x0         | reserved                                                                                                                                                                                                                                                                                                                                                                            |

**GRF\_IO\_VSEL**

Address: Operational Base + offset (0x0e640)

| Bit   | Attr | Reset Value | Description                                                                                                                                                                                                                                                                                                                                                                         |
|-------|------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | RW   | 0x0000      | write_enable<br>bit0~15 write enable<br>When bit 16=1, bit 0 can be written by software .<br>When bit 16=0, bit 0 cannot be written by software;<br>When bit 17=1, bit 1 can be written by software .<br>When bit 17=0, bit 1 cannot be written by software;<br>.....<br>When bit 31=1, bit 15 can be written by software .<br>When bit 31=0, bit 15 cannot be written by software; |
| 15:4  | RO   | 0x0         | reserved                                                                                                                                                                                                                                                                                                                                                                            |
| 3     | RW   | 0x0         | gpio1833_gpio4cd_ms                                                                                                                                                                                                                                                                                                                                                                 |
| 2     | RW   | 0x0         | sdmmc_gpio4b_ms                                                                                                                                                                                                                                                                                                                                                                     |
| 1     | RW   | 0x0         | audio_gpio3d4a_ms                                                                                                                                                                                                                                                                                                                                                                   |
| 0     | RW   | 0x0         | bt656_gpio2ab_ms                                                                                                                                                                                                                                                                                                                                                                    |

# io -r -4 0xff320180

ff320180: 00000300

# io -r -4 0xff77e640

ff77e640: 00000002